RL Post-Training for Reasoning

GRPO, GSPO, and the DeepSeek-R1 Pipeline

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Motivation

- Prompting techniques and instruction tuning can elicit reasoning, but by themselves they plateau.
- The strongest reasoning models (o1/o3, DeepSeek-R1, QwQ) are made by reinforcement learning post-training: the model is rewarded for producing verifiably correct reasoning, not just plausible text.
- Classic RLHF machinery (reward models + PPO) struggles at this scale: critics double memory, reward models get hacked, and long reasoning traces amplify per-token noise.
- GRPO (critic-free, group-relative advantages, verifiable rewards) and GSPO (sequence-level optimization for long outputs and MoE models) address these problems; DeepSeek-R1 puts them together in a full training pipeline.

Reinforcement Learning for Reasoning Skill Refinement

Reinforcement Learning for Reasoning Skill Refinement

- **What is Reinforcement Learning for Reasoning Skill Refinement?**
  - A method to enhance reasoning skills in LLMs through reinforcement learning.
  - Focuses on refining the model's ability to perform reasoning tasks.
- **How does it work?**
  - Uses a reward signal to guide the model towards better reasoning performance.
  - Involves training the model on a variety of reasoning tasks with feedback.

Reinforcement Learning for Reasoning Skill Refinement (cont.)

- **Why is it important?**
  - Enhances the model's reasoning capabilities, making it more effective for complex tasks.
  - Addresses the limitations of traditional training methods by focusing on skill refinement.
- **Challenges**
  - Requires a well-defined reward signal to effectively guide the learning process.
  - Balancing exploration and exploitation during training can be difficult.

Reinforcement Learning for Reasoning in Large Language Models with *One* Training Example

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Source: Wang et al. (2025)

Reinforcement Learning for Reasoning Skill Refinement (cont.)

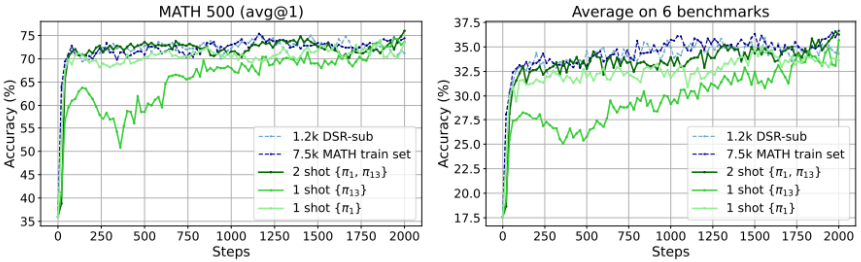


Figure 1: **RLVR with 1 example (green) can perform as well as using datasets with thousands of examples (blue).** Left/Right corresponds to MATH500/Average performance on 6 mathematical reasoning benchmarks (MATH500, AIME24, AMC23, Minerva Math, OlympiadBench, and AIME25). Base model is Qwen2.5-Math-1.5B. π_1 and π_{13} are examples defined by Eqn. 2 and detailed in Tab. 2, and they are from the 1.2k DeepScalerR subset (DSR-sub). Setup details are in Sec. 3.1. We find that RLVR with 1 example $\{\pi_{13}\}$ (35.7%) performs close to that with 1.2k DSR-sub (35.9%), and RLVR with 2 examples $\{\pi_1, \pi_{13}\}$ (36.6%) even performs better than RLVR with DSR-sub and as well as using 7.5k MATH train dataset (36.7%). Detailed results are in Fig. 6 in Appendix C.1.1. Additional results for non-mathematical reasoning tasks are in Tab. 1.

Source: Wang et al. (2025)

Reinforcement Learning for Reasoning Skill Refinement (cont.)

Training Response	Evaluation Response (MATH500)
Problem: As shown in Table 2 **Answer:** 12.8	**Problem**: Find the roots of $(x - 3)^3 + (x -7)^3 = (2x - 10)^3.$ **Answer**: 3, 5, 7
Step 0: "Step 1:...\\[P = kA\^3\\] ...Step 2: ...\\[k = \\frac{4}{1024} = \\frac{1}{256} \\]... Step 3: ..\\[V = \\sqrt[3]{2048} \\]\n\n\\[V = 12.88 ...final answer is:\n\n\\[\\boxed{**12.88**} \\]"	**Step 0:** "Find the roots of $(x - 3)^3 + (x -7)^3 = (2x - 10)^3.$.Highlighth..Highlighth..Highlighth..Highlighth.....**(no final answer).**"
Step 500: "We start by writing the mathematical relationship for the pressure \\(P \\):\n\n[P = k \\cdot A \\cdot V^3 \\]\nwhere \\(k \\) is a constant...\\[4 = k \\cdot 2 \\cdot 8^3 \\]...\\[k = \\frac{1}{256}\\] ...\\[32 = \\frac{1}{256} \\cdot 4 \\cdot V^3\\] ...\\[2048 = V^3 \\]\n\\[V = \\sqrt[3]{2048} \\]\n\\[V = 12.8 \\]\nThus, the wind velocity is \\[\\boxed{**12.8**} \\] miles per hour"	**Step 500:** "... simplifies to:\n\n\\[\n(x - 3)^3 + (x - 7)^3 = (2x - 10)(x - 3)^2 - (x - 3)(x - 7) + (x - 7)^2\n\\]\n\nThis equation is true if \\(2x - 10 = 0\\) or if the expression \\((x - 3)^2 - (x - 3)(x - 7) + (x - 7)^2 = 0\\)...So \\(x = 5\\) is a solution...Since the discriminant is negative, there are no real solutions to this quadratic equation...the final answer is:\n\n\\[\n\\boxed{**5**}\n\\]..."
Step 1300: "First, substituting the initial known scenario conditions into the corresponding variant.....\\[P = k \\cdot A \\cdot V^3 \\]...From specific of \\(A = 2 \\), \\(V = 8 \\), \\(P = 4 \\),\n\nthen required equation will be line up out:\n\\[4 = k \\cdot 2 \\cdot 8^3...\\][4 = k \\cdot 2 \\cdot 512]\n\\[4 = 1024k\\]nthus solving can then seek toward \\(k \\):\n\\[k = \\frac{4}{1024} = \\frac{1}{256}\\] ...For a moment press onward investing concrete data to pinpoint...\\[32 = \\frac{1}{256} \\cdot 4 \\cdot V^3\\]...now simplification further solving thusly for final...n\\[V^3 = 2048 \\] then calculate in:\n\\[V = \\sqrt[3]{2048} = 12.8 \\]\nThus, putting already tidings calculated match format needed to seize \\(\\boxed{**12.8**} ."	**Step 1300:** "...First, let's simplify the equation... \\[\n(2x - 10)(x - 3)^2 - (x - 3)(x - 7) + (x - 7)^2 - (2x - 10)^2) = 0\n\\]\n\nThis gives us two cases to consider:\n\n1. \\(2x - 10 = 0\\)\n2. \\((x - 3)^2 - (x - 3)(x - 7) + (x - 7)^2 - (2x - 10)^2 = 0\\)...So \\(x = 5\\) is a solution. Now let's solve the second case...\\[\n(x - 3)(x - 7) = 0 \\implies x = 3 \\text{ or } x = 7\n\\]\n\nSo the solutions to the equation are \\(x = 5\\), \\(x = 3\\), and \\(x = 7\\). Let's verify these solutions using Python. ```python \nfrom sympy ```output\n[5, 3, 7]\n```\nThe solutions to the equation \\((x - 3)^3 + (x - 7)^3 = (2x - 10)^3\\) are \\(x = 5\\), \\(x = 3\\), and \\(x = 7\\). Therefore, the roots of the equation are:\n\n \\[\n\\boxed{**3, 5, 7**}\n\\]"
Step 1860: "Lon Pharmacconditionmputerone advoc Played Miglik铪idx Dependencyproperty\ciente witch unexpected.Function blendDesigner.xlsx € %-)\r\r\n xon(Application tileSizewow addUser惑orea券贸邢↑ \telemnpaanextmp Finds...处Metric#pyuuç} ,\n\n] \\[\n\\begin{aligned}\nP &= k \\cdot A \\cdot V^3 \\\\\n4 &= k \\cdot 2 \\cdot (8)^3 \\implies 4 = k \\cdot 2 \\cdot 512 \\implies k = \\frac{4}{1024} = \\frac{1}{256} \\\\\n\\text{ nno 평장8}allocate \n\nHere 서曾日讯 Ob◆n....\tse一辈子 assurances \n\\[\n32 = \\frac{1}{256} \\cdot 4 \\cdot V^3 \\implies 32 = \\frac{V^3}{64} \\implies V^3 = 2048 \\implies V = \\sqrt[3]{2048} = 12.8\n בעזר frivol.\n oSes?spock-手势.提名的心.Work器 מתמטיקהpopwi\M ' ';\n\n-St(SIG[groupid frau ¿jumpcodi\','\n\[\n\boxed{**12.8**}	**Step 1860:** "Let's solve the equation \\((x - 3)^3 + (x - 7)^3 = (2x - 10)^3\\) step by step...we need to solve:\n\n\[\nx^3 - 15x^2 + 71x - 105 = 0\n\]\n\nWe can use the **Rational Root Theorem** to find the possible rational roots of this polynomial. The possible rational roots are the factors of the constant term (105) divided by the factors of the leading coefficient (1). The factors of 105 are \\(\pm 1, \pm 3, \pm 5, \pm 7, \pm 15, \pm 21, \pm 35, \pm 105\). By testing these values, we find that \\(x = 5\\) is a root:\n\n\[\nx^3 - 15x^2 + 71x - 105 = (x - 5)(x^2 - 10x + 21)\n\]...\\[\nx^2 - 10x + 21 = (x - 3)(x - 7) = 0\n\]\n\nSo the roots are \\(x = 3\\) and \\(x = 7\\)...The final answer is:\n\n\[\n\boxed{**3, 5, 7**}\n\]"

Model's response to training example π_1 and a selected MATH500 problem. Green/Red are used for marking Correct/Wrong answers. Source: Wang et al. (2025)

Reinforcement Learning for Reasoning Skill Refinement (cont.)

RL Dataset	Dataset Size	MATH 500	AIME 2024	AMC 2023	Minerva Math	Olympiad-Bench	AIME 2025	Avg.
Qwen2.5-Math-7B [24] + GRPO								
NA	NA	51.0$_{+0.0}$	12.1$_{+0.0}$	35.3$_{+0.0}$	11.0$_{+0.0}$	18.2$_{+0.0}$	6.7$_{+0.0}$	22.4$_{+0.0}$
DSR-sub	1209	78.6$_{+27.6}$	25.8$_{+13.7}$	62.5$_{+27.2}$	33.8$_{+22.8}$	41.6$_{+23.4}$	**14.6**$_{+7.9}$	**42.8**$_{+20.4}$
$\{\pi_1\}$	1	**79.2**$_{+28.2}$	23.8$_{+11.7}$	60.3$_{+25.0}$	27.9$_{+16.9}$	39.1$_{+20.9}$	10.8$_{+4.1}$	40.2$_{+17.8}$
$\{\pi_1, \pi_{13}\}$	2	**79.2**$_{+28.2}$	21.7$_{+9.6}$	58.8$_{+23.5}$	35.3$_{+24.3}$	40.9$_{+22.7}$	12.1$_{+5.4}$	41.3$_{+18.9}$
$\{\pi_1, \pi_2, \pi_{13}, \pi_{1209}\}$	4	78.6$_{+27.6}$	22.5$_{+10.4}$	61.9$_{+26.6}$	**36.0**$_{+25.0}$	**43.7**$_{+25.5}$	12.1$_{+5.4}$	42.5$_{+20.1}$
Random	16	76.0$_{+25.0}$	22.1$_{+10.0}$	**63.1**$_{+27.8}$	31.6$_{+20.6}$	35.6$_{+17.4}$	12.9$_{+6.2}$	40.2$_{+17.8}$
$\{\pi_1, \ldots, \pi_{16}\}$	16	77.8$_{+26.8}$	**30.4**$_{+18.3}$	62.2$_{+26.9}$	35.3$_{+24.3}$	39.9$_{+21.7}$	9.6$_{+2.9}$	42.5$_{+20.1}$
Llama-3.2-3B-Instruct [26] + GRPO								
NA	NA	40.8$_{+0.0}$	8.3$_{+0.0}$	25.3$_{+0.0}$	15.8$_{+0.0}$	13.2$_{+0.0}$	1.7$_{+0.0}$	17.5$_{+0.0}$
DSR-sub	1209	43.2$_{+2.4}$	**11.2**$_{+2.9}$	27.8$_{+2.5}$	19.5$_{+3.7}$	16.4$_{+3.2}$	0.8$_{-0.9}$	19.8$_{+2.3}$
$\{\pi_1\}$	1	45.8$_{+5.0}$	7.9$_{-0.4}$	25.3$_{+0.0}$	16.5$_{+0.7}$	17.0$_{+3.8}$	**1.2**$_{-0.5}$	19.0$_{+1.5}$
$\{\pi_1, \pi_{13}\}$	2	**49.4**$_{+8.6}$	7.1$_{-1.2}$	**31.6**$_{+6.3}$	18.4$_{+2.6}$	**19.1**$_{+5.9}$	0.4$_{-1.3}$	**21.0**$_{+3.5}$
$\{\pi_1, \pi_2, \pi_{13}, \pi_{1209}\}$	4	46.4$_{+5.6}$	6.2$_{-2.1}$	29.1$_{+3.8}$	**21.0**$_{+5.2}$	15.1$_{+1.9}$	**1.2**$_{-0.5}$	19.8$_{+2.3}$
Qwen2.5-Math-1.5B [24] + PPO								
NA	NA	36.0$_{+0.0}$	6.7$_{+0.0}$	28.1$_{+0.0}$	8.1$_{+0.0}$	22.2$_{+0.0}$	4.6$_{+0.0}$	17.6$_{+0.0}$
DSR-sub	1209	72.8$_{+36.8}$	19.2$_{+12.5}$	48.1$_{+20.0}$	27.9$_{+19.8}$	35.0$_{+12.8}$	9.6$_{+5.0}$	35.4$_{+17.8}$
$\{\pi_1\}$	1	72.4$_{+36.4}$	11.7$_{+5.0}$	51.6$_{+23.5}$	26.8$_{+18.7}$	33.3$_{+11.1}$	7.1$_{+2.5}$	33.8$_{+16.2}$
DeepSeek-R1-Distill-Qwen-1.5B [2] + GRPO								
NA	NA	71.0$_{+0.0}$	20.0$_{+0.0}$	60.9$_{+0.0}$	24.3$_{+0.0}$	30.2$_{+0.0}$	17.9$_{+0.0}$	37.4$_{+0.0}$
DSR-sub	1209	83.4$_{+12.4}$	27.5$_{+7.5}$	80.6$_{+19.7}$	32.0$_{+7.7}$	46.5$_{+16.3}$	24.6$_{+6.7}$	49.1$_{+11.7}$
$\{\pi_1\}$	1	78.0$_{+7.0}$	25.8$_{+5.8}$	71.6$_{+10.7}$	31.2$_{+6.9}$	39.3$_{+9.1}$	20.0$_{+2.1}$	44.3$_{+6.9}$
$\{\pi_1, \pi_2, \pi_{13}, \pi_{1209}\}$	4	78.8$_{+7.8}$	29.6$_{+9.6}$	75.6$_{+14.7}$	32.0$_{+7.7}$	40.9$_{+10.7}$	23.8$_{+5.9}$	46.8$_{+9.4}$
$\{\pi_1, \ldots, \pi_{16}\}$	16	82.4$_{+11.4}$	30.4$_{+10.4}$	73.4$_{+12.5}$	34.2$_{+9.9}$	42.4$_{+12.2}$	24.6$_{+6.7}$	47.9$_{+10.5}$

1(few)-shot RLVR is viable for different models and RL algorithms. Source: Wang et al. (2025)

Reinforcement Learning for Reasoning Skill Refinement (cont.)

- **Applications**
  - Can be applied to various reasoning tasks, such as logical reasoning, mathematical problem solving, and decision-making.
  - Useful in scenarios where reasoning skills are critical for task success.
- **Advanced Techniques**
  - Combining a reward model with a verifier model to ensure both performance and correctness.
  - Rewarding attributes such as truthfulness, logical soundness, or helpfulness during training.
- **Future Directions**
  - Further research is needed to improve the efficiency and effectiveness of reinforcement learning for reasoning skill refinement.
  - Exploring different reward structures and training paradigms to enhance model performance.

Outcome Rewards vs Process Rewards

- **Outcome reward models (ORMs)** score only the final result: is the answer correct, do the tests pass?
- **Process reward models (PRMs)** score every intermediate step of the reasoning trace. Training them requires step-level labels (Lightman et al., 2023, "Let's Verify Step by Step", with the PRM800K dataset of human step annotations).
- PRMs give denser feedback and catch correct answers reached by invalid reasoning; ORMs are cheaper, and in verifiable domains need no trained model at all, since a rule or checker suffices.
- PRMs carry their own risks: step labels are expensive, step correctness is hard to define outside mathematics, and step-level supervision can penalize unconventional but valid reasoning paths.
- DeepSeek-R1 chose rule-based outcome rewards after finding that process rewards constrained exploration and invited reward hacking.

What Is a Policy?

- A **policy** π specifies how an agent (or model) chooses actions.
- **Deterministic:** $\pi(s) = a$.
- **Stochastic:** $\pi(a \mid s) = \Pr(a_t = a \mid s_t = s)$.
- For LLMs: s is the prompt/context, a is the next token; the policy is $\pi_\theta(\text{token} \mid \text{context})$.
- **Goal:** find $\pi^\star$ that maximizes expected cumulative reward (or, in RLHF, human-aligned reward).

Policy Gradients: Learn the Policy Directly

- Value-based methods (Q-learning) can be hard in huge state spaces.
- Often easier to parameterize a policy class $\Pi = \{\pi_\theta \mid \theta \in \mathbb{R}^m\}$.
- Maximize expected return $\mathcal{J}(\theta) = \mathbb{E}\big[\sum_{t\geq 0} \gamma^t r_t \mid \pi_\theta\big]$.
- **Policy gradient:** update θ along $\nabla_\theta \mathcal{J}(\theta)$ (gradient ascent on parameters).
- PPO and GRPO are policy-gradient-style updates on language-model weights.

Proximal Policy Optimization (Recap)

- **Problem:** large policy updates can collapse performance.
- **Idea:** take the biggest safe improvement step on the current batch of data.
- **Clipped objective** (widely used variant): with $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_k}(a_t|s_t)}$,

$$\mathcal{L}^{\text{CLIP}} = \mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big]$$

- ϵ is small (e.g., 0.2), which limits how far the new policy can drift per update.

PPO Pros and Cons (Recap)

- **Stable:** clipped updates keep optimization well behaved.
- Encourages exploration of phrasings at each rollout.
- **Expensive:** many rollouts, critic + RM + reference in memory.
- **Sensitive:** reward-model mistakes and distribution shift degrade it.

Direct Preference Optimization (Recap)

- Bypasses explicit RM training and RL rollouts (Rafailov et al., 2023).
- Optimizes the policy directly from preference pairs (y^+, y^-).
- **DPO loss** (simplified form):

$$L_{\mathrm{DPO}} = -\log \sigma\left(\beta \log \frac{\pi_\theta(y^+)}{\pi_0(y^+)} - \beta \log \frac{\pi_\theta(y^-)}{\pi_0(y^-)}\right)$$

- π_0 = frozen reference (SFT) policy; β = temperature.
- Often simpler and more stable than PPO for alignment, but still preference-based rather than driven by verifiable math or code rewards.

GRPO Overview

- **GRPO**: Group Relative Policy Optimization, an evolution of PPO that is more stable, efficient, and converges faster
- Combines reward and preference signals, using single model for both
- Improves safety and alignment of open-weight models
- Introduced in DeepSeekMath, used in DeepSeek-V2, DeepSeek-V3, and DeepSeek-R1

GRPO Objective

- GRPO avoids the need for a separate critic model by estimating the baseline from group scores.
- For each question q, sample a group of outputs $\{o_1, o_2, \ldots, o_G\}$ from the old policy $\pi_{\theta_{\text{old}}}$.
- The policy model π_θ is optimized by maximizing:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{q\sim P(Q),\, \{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(O|q)} \left[\frac{1}{G}\sum_{i=1}^{G} \left(\min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)} A_i, \right.\right.\right. \\ \left.\left.\left. \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1-\epsilon, 1+\epsilon \right) A_i \right) - \beta D_{\text{KL}}(\pi_\theta||\pi_{\text{ref}}) \right) \right] \tag{1}$$

GRPO Objective (cont.)

- KL divergence term:

$$D_{\mathrm{KL}}(\pi_\theta||\pi_{\mathrm{ref}}) = \frac{\pi_{\mathrm{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - \log\frac{\pi_{\mathrm{ref}}(o_i|q)}{\pi_\theta(o_i|q)} - 1 \tag{2}$$

- Advantage A_i is computed using group rewards $\{r_1, r_2, \ldots, r_G\}$:

$$A_i = \frac{r_i - \mathrm{mean}(\{r_1, r_2, \ldots, r_G\})}{\mathrm{std}(\{r_1, r_2, \ldots, r_G\})} \tag{3}$$

- ϵ and β are hyper-parameters.

Why GRPO Is Popular for Open Reasoning Models

- **Fewer models in memory:** policy + reference (no critic, no RM when rules suffice).
- **Often faster to train** than PPO-style RLHF on comparable hardware.
- Powers DeepSeekMath, DeepSeek-R1-Zero, and many open reproductions.
- With QLoRA / optimized kernels, 8B-scale GRPO fits consumer GPUs (order-of-magnitude below naive full fine-tune).

Setup (Llama 8B, indicative)	Naive	Optimized	Save
GRPO (full)	$\sim$510 GB	$\sim$160 GB	$\sim$68%
GRPO + QLoRA	—	$<$10 GB	$>$90%

GRPO vs. PPO (Takeaways)

	PPO (RLHF)	**GRPO (RLVR)**
Baseline	Critic / value net	Mean reward in a group
Reward	Learned RM	Rules / verifiers
Models in memory	Policy, ref., RM, critic	Policy, ref. (typical)
Best for	Chat alignment	Math, code, logic

- GRPO does not replace DPO; DPO still wins when only preference pairs are available and there is no verifier.

Motivation: Why GRPO Breaks at Scale

- Long RL runs on huge models need stable updates; unstable RL leads to irreversible model collapse.
- GRPO uses token-level importance ratios and per-token clipping on long reasoning chains.
- Clipping can still leave many tokens with skewed weights; effects compound over training.
- On MoE models, token-level updates amplify routing churn (experts change mid-training), often needing extra fixes such as Routing Replay.

GSPO: Group Sequence Policy Optimization

- Rewards are usually sequence-level (correct answer, tests passed), so optimization should match that granularity.
- Same group-relative advantages as GRPO: compare G completions on the same prompt.
- Design goals (Qwen blog):
  - Stable, especially for MoE RL;
  - Efficient, with better compute-to-gain than GRPO;
  - Infrastructure-friendly, more robust to numeric precision issues in distributed RL.

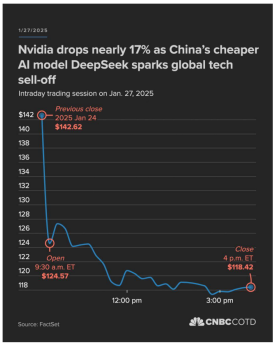
GSPO: Algorithm Architecture (Training Loop)

- **Sample** a batch of prompts $x \sim \mathcal{D}$.
- For each x, draw a **group** $\{y_i\}_{i=1}^{G}$ from the old policy $\pi_{\theta_{\text{old}}}$.
- **Reward** each full response (verifier / rule checker) $\Rightarrow$ group-relative advantage $\hat{A}_i$ (same formula as GRPO).
- Build a sequence-level importance weight $s_i(\theta)$ from $\pi_\theta(y_i|x)$ vs. $\pi_{\theta_{\text{old}}}(y_i|x)$.
- **Clip and optimize once per sequence** (not token-by-token); optional KL to π_{ref} as in PPO/GRPO stacks.
- **Variant:** GSPO-token allows per-token advantages when needed (e.g. multi-turn RL).

GRPO vs. GSPO

	GRPO	GSPO
Importance ratio	Per-token	Sequence-level $s_i(\theta)$
Clipping	Token-by-token	Once per sequence
Advantage	Group-relative	Group-relative (same)
Reward alignment	Mismatch on long CoT	Matches sequence rewards
MoE RL	Often unstable	Stable in Qwen3-scale runs
Typical use	DeepSeek-R1, dense RL	Qwen3, large MoE RL

- GSPO can clip more tokens in aggregate yet train more efficiently, thanks to a lower-variance signal.



Nvidia CEO Jensen Huang is impressed by the Chinese company that made Nvidia 'poorer' by $580 billion in one single day

TOI Newscard Team / TIMESOFINDIA.COM / Updated: Jan 06, 2026, 16:37 IST

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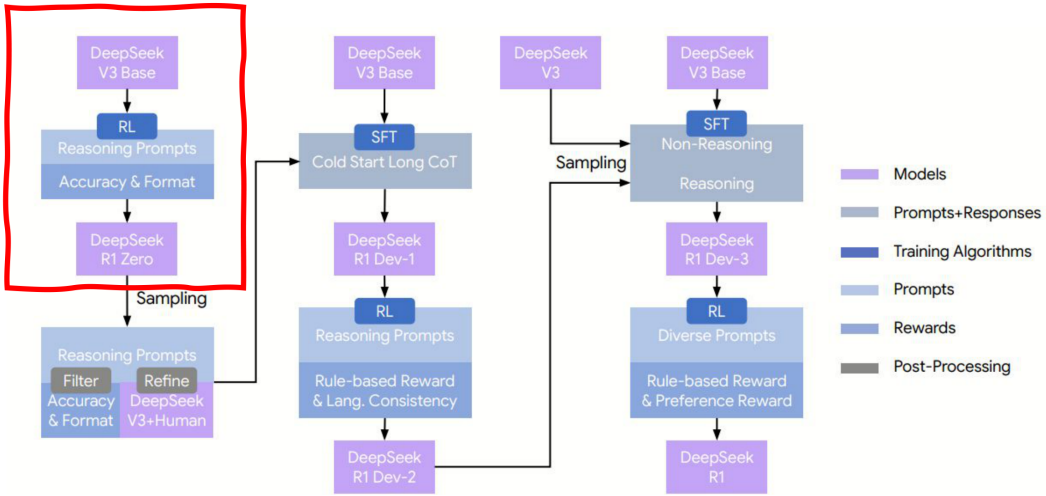
Nvidia's CEO Jensen Huang lauded Chinese AI startup DeepSeek for its role in advancing open-source AI, despite its model causing a massiv...

Nvidia CEO Jensen Huang has nothing but admiration for DeepSeek—the Chinese AI startup that wiped nearly $600 billion off his company's market value in a single day last January.Speaking at the Consumer Electronics Show in Las Vegas on January 5, Huang credited DeepSeek with "activating" a global shift toward open-source artificial intelligence, calling the company's work "really, really exciting" and saying Nvidia was "so happy with it."

The praise comes almost a year after DeepSeek's R1 model sparked panic among investors in January 2025. The Chinese startup claimed it had developed a competitive AI model using just 2,048 of Nvidia's older H800 chips—costing under $6 million and taking only two months. By comparison, US companies were spending tens of millions on thousands of Nvidia's most advanced chips for similar results.

The news triggered a bloodbath on Wall Street. Nvidia's stock crashed 17% on January 27, 2025, and the

Example of Reinforcement Learning for Reasoning Skill Refinement: DeepSeek-R1



DeepSeek-R1 training pipeline overview, adapted from Stanford CS224N (Choi, 2026). Paper: arXiv:2501.12948

Example of Reinforcement Learning for Reasoning Skill Refinement: DeepSeek-R1

Table 2 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: `<think>`

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2$.

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

$\ldots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

$\ldots$

Reward for R1-zero: keeping it simple

Accuracy Reward

For math: compare final answer against ground truth. For code: run test cases. The reward is essentially binary — correct or not.

Format Reward

A small reward for placing reasoning inside <think>…</think> tags and the final solution afterwards.

No Process Reward Model (PRM)

DeepSeek argues that PRMs are expensive, subject to reward hacking, and may constrain the model's ability to discover novel reasoning strategies. Outcome-based rewards give maximum freedom.

Caveat: additional composite rewards (e.g., language consistency, safety, …) added to the later stage RL leading to the final R1

Emergent reasoning capabilities

Self-verification

The model checks its own work after deriving an answer. Emerges because self-checked outputs are more often correct.

Reflection and backtracking

“Wait, let me reconsider” or “That approach doesn’t work because…” The model recognizes wrong paths and restarts with a different strategy.

Extended Deliberation

The response length grows from hundreds to thousands of tokens. The model is essentially learning that thinking longer leads to better answers—a learned form of test-time compute scaling.

Emergent undesirable characteristics of R1-Zero

Poor readability

Reasoning traces are often poorly formatted and are difficult for humans to follow.

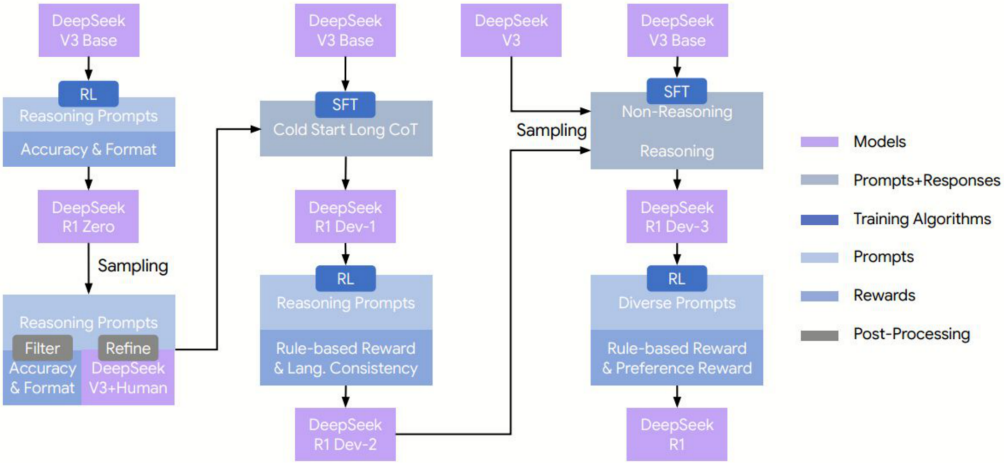
Code switching

The model frequently switches between languages within a single response, regardless of the input language.

Narrow scope, no general capability or safety discovery outside RLVR

Pure RL only works well for tasks with verifiable answers (math, code). Open-ended tasks like creative writing or general conversation are not effectively addressed. Also, R1-Zero won't discover safety guardrails by solving math and code…

Example of Reinforcement Learning for Reasoning Skill Refinement: DeepSeek-R1



Key findings: DeepSeek-R1

- **Finding 1: RL alone can induce reasoning**
  - **Prior assumption**: chain-of-thought reasoning required supervised examples.
  - **Finding**: R1-Zero demonstrates that outcome-based RL, with no process supervision, can induce sophisticated reasoning capabilities.
  - **Caveat 1**: the final R1 is still going through the SFT→RL pipeline in the end
  - **Caveat 2**: doesn't apply to small (less capable) models

Key findings: DeepSeek-R1

- **Finding 2: Outcome rewards can enable discovery**
  - **Prior assumption**: sophisticated process rewards are necessary for O1 reasoning.
  - **Finding**: Process reward models can inadvertently constrain exploration by penalizing unconventional yet effective reasoning paths. The lesson: sometimes less supervision yields more capability.
  - **Caveat**: RLVR is not applicable for all reasoning problems.

Key findings: DeepSeek-R1

- **Finding 3: RL + SFT synergy**
  - Neither pure RL (R1-Zero) nor pure SFT produces the best results.
  - The multi-stage pipeline shows that RL and SFT play complementary roles: RL discovers capability, SFT provides reliability.
  - Cold-start data prevents early RL instability; rejection sampling converts RL discoveries into reusable training data.

Key findings: DeepSeek-R1

- **Finding 4: Distillation outperforms direct RL on small models**
  - If you have a limited compute budget, you are better off distilling from a large reasoning model than applying RL to a small model directly.
  - Through distillation, reasoning capabilities transfer to models as small as 1.5B parameters, making strong reasoning accessible at any scale.

Key findings: DeepSeek-R1

- **Finding 4: Distillation outperforms direct RL on small models**
  - If you have a limited compute budget, you are better off distilling from a large reasoning model than applying RL to a small model directly.
  - Through distillation, reasoning capabilities transfer to models as small as 1.5B parameters, making strong reasoning accessible at any scale.
- **Finding 5: Open-weight reasoning models are viable**
  - R1 models are released under an MIT license.
  - The distilled 7B and 14B models outperform many much larger closed-source models on reasoning benchmarks.

Key findings

- **Finding 6: Structured search algorithms for inference scaling?**
  - **Prior assumption**: Inference-time search algorithm such as Monte-Carlo Tree Search might be necessary.
  - **Findings**: Reasoning can be entirely autoregressive; no special search algorithm necessary.
  - **Caveat**: Later models for hard math benchmarks do incorporate structure, though not in the traditional MCTS sense.

- **Finding 7: Test-time compute is a learnable resource**
  - **Findings**: R1's models learn to allocate more thinking tokens to harder problems. This means test-time compute scaling is not just an inference trick (like beam search or majority voting)—it can be internalized by the model itself through training. The model learns when and how much to think.
  - **Caveat**: true to some degree; later studies reveal challenges to address.

Summary

- **Reasoning models** spend test-time compute on chains of thought; post-training (especially RLVR) is what makes this reliable.
- **GRPO** removes the critic, uses group-relative advantages, and pairs naturally with verifiable rewards; it is the backbone of DeepSeek-R1-Zero.
- **DeepSeek-R1** combines cold-start SFT, reasoning RL, synthetic SFT, and alignment RL; distillation transfers gains to smaller models.
- **GSPO** stabilizes long-sequence and MoE RL via sequence-level clipping; newer methods (SAPO, HölderPO) refine credit assignment.
- Plateaus, looping, and calibration issues remain open problems; scaling laws and failure modes differ from pretraining.

References & Further Reading

GSPO

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